

$$\begin{aligned}
A_\tau &\triangleq \{a \in A \mid \text{tid}(a) = \tau\} \text{ and } r_\tau \triangleq r \cap (A_\tau \times A_\tau) \\
r_i &\text{ for internal (same-thread) } r \text{ edges:} \\
r_i &\triangleq \{(a, b) \in r \mid \text{tid}(a) = \text{tid}(b) \vee \text{tid}(a) = 0\} \\
r_e &\text{ for external } r \text{ edges: } r_e \triangleq r \setminus r_i \\
A_x &\triangleq \{a \in A \mid \text{loc}(a) = x\} \text{ and } r_x \triangleq r \cap (A_x \times A_x) \\
r|_{\text{loc}} &\triangleq \{(a, b) \in r \mid \text{loc}(a) = \text{loc}(b)\} \\
G.E &\triangleq E \\
G.po &\triangleq po \\
G.\text{rf} &\triangleq rf \\
G.R_x &\triangleq G.R \cap \{r \in E \mid \text{loc}(r) = x\}
\end{aligned}$$